

CH2 Combinatorial Methods

2.2 Counting principle

Theorem 2.1 (Counting Principle)

If the set E contains n elements and the set F contains m elements, then, there are $n \cdot m$ ways in which we can choose

Theorem 2.2 (Generalized Counting Principle)

If we have sets $E_1, E_2, E_3, \dots, E_n$

then, there are $n_1 \times n_2 \times n_3 \times \dots \times n_n$ ways to choose

Example 2.2

In tossing 4 fair dice, $P(\text{at least one 3 among these 4 dice}) = ?$

$$N(S) = 6 \times 6 \times 6 \times 6 = 6^4$$

$$N(A^C) = 5 \times 5 \times 5 \times 5 = 5^4$$

Example 2.6 (Standard Birthday Problem)

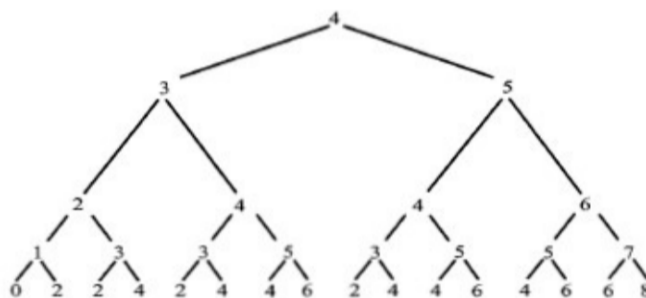
$P(\text{at least two among } n \text{ people have the same birthdays}) =$

$$N(S) = 365 \times 365 \times \dots \times 365 = 365^n$$

$$N(A^C) = 365 \times 364 \times \dots \times (365 - n + 1)$$

Counting principle: Tree Diagram

Ex 2.9 Mark has \$4. He decides to bet \$1 on the flip of a fair coin 4 times. What is the probability that (a) he breaks even; (b) he wins money
Ans: (a) 6/16 (b) 5/16



2.3 Permutations

An **ordered** arrangement of r object in set A containing n object is called:

r element permutation of A denote as ${}_nP_r$, $P(n, r)$, P_r^n

$$P_r^n = n \times (n-1) \times (n-2) \times \dots \times (n-r)$$

Ex 2.11: 2 electronics, 4 computer science, 3 statistics, 3 biology, and 5 music books are put on a bookshelf with a random arrangement. What is the probability the books of the same subject are together?

→ 分組後在組內排序

$$N(A) = 5! \times (2! \times 4! \times 3! \times 3! \times 5!)$$

分成五個組別 每個組別內排序

Theorem 2.4

The number of distinguishable permutations of n objects with k different types

$$\frac{n!}{n_1! \times n_2! \times \dots \times n_k!}$$

With replacement: 可以重複 (Sample space 不會變小)

Ex 2.15 A fair coin is flipped 10 times. $P(\text{exactly 3 heads}) =$

$$N(S) = 2^{10}; N(A) = 10! / (3! * 7!)$$

2.4 Combinations

Definition

An **unordered** arrangement of r objects from a set of A containing n objects is called an r -element **combination** of A , denote as ${}_nC_r, C(n, r) C_r^n$

A Few Notes

$$C_0^n = C_n^n = 1$$

$$C_1^n = C_{n-1}^n = n$$

$$C_r^n = C_{n-r}^n \quad (10 \text{ 個人, 選3放室內與選 7 放室外 是一樣的})$$

$$C_r^{n+1} = C_r^n + C_{r-1}^n \quad (\text{設有 1 藍球與 9 白球, 選 3 顆 的情況 } C_3^{10} \text{ 可以是})$$

→ 沒有包含藍球 C_3^9 加上確定有藍球 C_2^9

Ex 2.17 45 instructors were selected randomly to ask whether they are happy with their teaching loads. The response of 32 were negative. If Drs. Smith, Brown, and Jones were among those questioned. $P(\text{all 3 gave negative responses})=?$

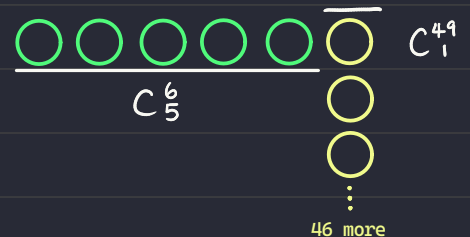
Ex 2.19 In the lottery game, players pick 6 integers from 1 to 49, order of selection being irrelevant.

(a) $P(\text{Grand prize} = 6 \text{ matches}) = ?$

$$P = 1 \div C_6^{49}$$

(b) $P(\text{2nd prize} = 5 \text{ matches}) = ?$

$$P = (C_5^6 \times C_1^{43}) \div C_6^{49}$$



Ex 2.20 7 cards are drawn from 52 without replacement.

$P(\text{at least one of the cards is a king})=?$

算 A^c ...

Ex 2.21 5 cards are drawn from 52. $P(\text{full house})=?$



Theorem (放隔板定理)

(a) The number of ways n indistinguishable objects can be placed into k distinguishable is

(b) The number of distinct nonnegative integer solutions that the equation

$x_1 + x_2 + x_3 + \dots + x_k = n$ has is

$$\binom{n+k-1}{n} = \binom{n+k-1}{k-1}$$

Theorem 2.5 (Binomial Theorem; Binomial expansion)(二項式定理)

$$(x+y)^n = \sum_{i=0}^n \binom{n}{i} x^{n-i} y^i$$

幾種該項的組合: $(x+y)^2 = x^2 + 2xy + y^2$

Example: What is the coefficient of the x^2y^3 in the expansion of $(2x+3y)^5$

$$\binom{5}{3}$$

Example: Evaluate the sum $\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n}$

將二項式定理 $x=1, y=1$ 代入 $\rightarrow 2^n$

Example: What is the coefficient of the x^3y^2 in the expansion of $(2x+3y)^5$

$$\sum_{i=0}^n (2x)^{n-i} (3y)^i$$

■ Ex 2.28 Prove that

$$\binom{2n}{n} = \sum_{i=0}^n \binom{n}{i}^2$$

Hint:

$$2n = n + n; \quad \binom{n}{i} = \binom{n}{n-i}$$

Theorem 2.6 (Multinomial expansion)

$$(x_1 + x_2 + \cdots + x_k)^n = \sum_{n_1 + n_2 + \cdots + n_k = n} \frac{n!}{n_1! n_2! \cdots n_k!} x_1^{n_1} x_2^{n_2} \cdots x_k^{n_k}$$

This theorem is from Theorem 2.4